

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 35.2

$$\begin{aligned} & \int 3^{4x} dx \\ & \stackrel{4x=u}{=} \int 3^u dx \\ & \stackrel{dx=\frac{du}{4}}{=} \int 3^u \cdot \frac{du}{4} \\ & = \frac{1}{4} \int 3^u du \\ & = \frac{1}{4} \cdot \frac{3^u}{\ln(3)} + C \\ & \stackrel{u=4x}{=} \frac{3^{4x}}{4 \ln(3)} + C \end{aligned}$$

References